



The Datsun redi-Go is one of the first cars to feature an auto 'box in the entry segment



The new Honda Amaze brings superior CVT tech to compact sedans

consumer behavior towards adopting the AMT technology in the entry level segment. Currently, Datsun redi-GO 1L AMT contributes approximately 25% to the overall sales of the Datsun redi-GO model. We foresee a healthy demand for AMT variants in this segment and specifically so for the Datsun redi-GO, owing to its ability to effectively meet customer expectations around easy to drive technology in stop and go city traffic. Nissan on the other hand, offers CVT (continuously variable transmission) technology in both Nissan Sunny and Micra in India. The CVT technology delivers more mileage than a manual transmission for added value in terms of fuel economy."

This itself signifies a shift towards basic automatic gearboxes in the entry-segments, and better technologies going forward. In these segments, while space, utilities and mileage would be the key elements that one usually looks for in India, a quick look at all the offerings in this budget segment reveals the changing nature of the industry.

Incidentally, the most price- and value-conscious segment of the Indian car market is also the biggest. With a perennial crunch for available space, hatchbacks make the most sense for Indian roads. While most Indian car buyers have been aware of this, it is only of late, with the availability of more sophisticated technology refining the AMT gearboxes, that a clear shift towards automatic gearboxes is being seen.

Better tech at heart

Saigot further elucidates the kind of technology that is steadily going into cars at the entry point with automatic gearboxes. He says, "(the Datsun redi-Go has) a two pedal technology and uses actuators for clutch and gear shift operations. Actuators are either electro-mechanical or electro-hydraulic type. The mechanism includes a clutch, but no clutch pedal. Clutch and Gear shifts are operated by actuators, and all this is precisely controlled by the transmission control unit (TCU)."

Tata Motors, too, is following in the same direction, with its new AMT technology in the Tata Nexon, proprietarily called 'HypDrive S-SG', featuring elements like anti-stall, fast launch, kick-down, crawl and hill-start assist to counter all the negative factors that AMT gearboxes were infamous for. Maruti Suzuki, too, has been refining its AMT gear-switching technology with the AGS gearboxes in cars. These include developed actuators that help a car to not stall in the middle of a journey or while shifting gears, particularly in the lower revs. Initial launch in AMT-equipped cars were also a major issue, and there was a perceivable lack of power and response from the engine because of sluggish gear shifts.

The crawl function, enabled by technology principles borrowed from superior, dual-clutch automatic gearboxes, is the biggest facilitator for the adoption of AMT-equipped cars. With

heavy city traffic, premium and luxury cars have always had the benefit of crawl, wherein the car can very slowly crawl along without the constant application of clutch and brakes. This feature is now gradually making its way to cars at the entry segments, and can make a world of difference, particularly in congested roads with slow-moving traffic.

In the slightly higher B and B2 segments, cars gradually feature CVT gearboxes. As opposed to AMTs, CVT 'boxes provide lesser shift-shock and provide much smoother gear shift curves, without any sudden drop in power between shifts.

Moving forward

As we slowly move towards electric mobility and maybe even one-pedal driving, the Indian car industry is gradually but clearly moving towards automatic gearboxes. As Saigot says, "AMT gearboxes have gained immense popularity in India in the recent years, and its use in affordable entry-level cars is more prevalent than before. With newer technologies, judicious pricing, improvements in mileage, and the promise of a smoother and less-stressful commute, things are changing—and quite rapidly."

Seeing how our roads aren't exactly a great pleasure to drive on for fans of the stick shift, the shift towards automatic gearboxes is the definite way forward in the next decade. ■